Answer 1 (Raft & Multi-Paxos):

Log Matching Property means if two logs have an entry with same term and index, they have the same command and identical previous entries.

In Raft, a leader cannot commit older entries directly by counting server responses.

The leader has to commit an entry from its own current term, which then commits older entries.

This prevents a partitioned leader from overwriting entries after reconnecting.

Comparison: In steady state, both Raft and Multi-Paxos take 1 RTT for log replication.

Leader failure: Raft election chooses a leader that already has all committed logs by checking log length and term during voting. In Multi-Paxos, any node can become leader and it has to run Phase 1 to discover what past leaders proposed, which is much more complicated.

Answer 2 (Concurrency & Snapshot Isolation):

Strict 2PL: Transactions have two phases. In Phase 1 prepare message is sent, in Phase 2 commit message.

[Note: Confused 2PL locking with 2-Phase Commit 2PC protocol!]

Rigorous 2PL holds locks until the end of the transaction.

Wait-For Graph is used to find deadlocks by looking for cycles between waiting transactions.

Write Skew in Snapshot Isolation: Suppose we need active doctors >= 1. Two doctors D1 and D2 are active.

Doctor 1 logs in and sees 2 doctors, so D1 goes off duty and commits.

Doctor 2 logs in concurrently, also sees 2 doctors active on snapshot, so D2 goes off duty.

Because D1 updated D1 and D2 updated D2, their write sets are different so both transactions succeed.

This leaves 0 doctors active, which breaks the consistency rule.

SSI prevents write skew by tracking read-write anti-dependencies and aborting one transaction.

Answer 3 (Consistent Hashing & Dynamo):

Consistent hashing puts keys on a circle. Virtual nodes give each physical server multiple tokens.

This makes key distribution balanced. When 1 node is added to N nodes, only 1/(N + 1) keys need to move.

Quorum condition: R + W > N. This ensures read and write sets overlap on at least 1 server.

Vector clocks track version numbers for each node to detect conflicts. LWW uses timestamps but clock drift causes data loss when writes are dropped.

Answer 4 (Failure Detectors & PBFT):

Failure detector <>W has accuracy property where some correct node is not suspected.

FLP impossibility says consensus cannot be solved in asynchronous systems with failures.

<>W helps solve it by adding timing assumptions.

PBFT requires N >= 3f + 1 because Byzantine nodes can send malicious messages and crash.

If f nodes crash, we need N - f to answer. If f nodes lie, we need N - 2f > f, so N >= 3f + 1.